INCH-POUND
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MILITARY SPECIFICATION

EJECTOR ASSEMBLIES, AIR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers air ejector assemblies to serve surface condensers for Naval shipboard application.

1.2 Classification. Air ejector assemblies shall be of the following types and classes, as specified (see 6.2 and 6.6).

Type I - For service of main, ship's service, or other auxiliary surface condensers.

Type II - For service of distilling unit condensers (surface ship and submarine non-submergence pressure applications).

Class 1 - Basically 70-30 copper-nickel on seawater side.
Class 2 - Basically 90-10 copper-nickel on seawater side.
Class 3 - Basically titanium tubes, tube sheets, and shell internals.

Type III - For service of condensers in a heating drain system or turbine gland leakoff system.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 05Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
TT-P-28	- Paint, Aluminum, Heat Resisting (1200°F).
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901	- Shock Tests, H I (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-A-907	- Antiseize Thread Compound, High Temperature.
MIL-S-1222	- Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts.
MIL-R-6855	- Rubber, Synthetic Sheets, Strips, Molded, or Extruded Shapes, General Specification for.
MIL-T-15005	- Tubes, 70-30 and 90-10 Copper Nickel Alloy, Condenser, and Heat Exchanger.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-C-15430	- Condensers, Steam, Naval Shipboard.
MIL-C-15726	- Copper-Nickel Alloy, Sheet, Plate Strip, Bar, Rod, and Wire.
MIL-E-15809	- Expander, Tube, Condenser, and Heat Exchangers.
MIL-T-17286	- Turbines and Gears, Shipboard Propulsion and Auxiliary Steam; Packaging of.
MIL-D-18641	- Distillation Units, Water: Steam or Flashed Vapor Operated, or Fresh Water Heated, Low Pressure, Naval Shipboard.
MIL-G-18997	- Gauge, Pressure, Dial Indicating.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-A-19521	- Anodes, Corrosion Preventive, Zinc, and Plugs, Zinc Anode Retaining: Design of and Installation in Shipboard Condensers and Heat Exchangers.
MIL-V-20065	- Valves, Angle, Pressure Relief, Naval Shipboard, for Steam Service.
MIL-C-20159	- Copper-Nickel Alloy Castings (UNS No. C76200 and C76400).
MIL-R-21252	- Rubber Sheet, Solid, Synthetic, Shipboard Water Evaporator Gasketing.

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MILITARY (Continued)

MIL-G-21610 - Gaskets, Heat-Exchanger, Various Cross Section Rings, Synthetic Rubber.
MIL-S-22473 - Sealing, Locking, and Retaining Compounds: (Single-Component).
MIL-S-22698 - Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural.
MIL-L-24131 - Lubricant, Colloidal Graphite in Isopropanol.
MIL-L-24478 - Lubricant, Molybdenum Disulfide in Isopropanol.
MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings For Seawater Service.
MIL-C-24615 - Castings, Nickel-Chromium-Molybdenum-Columbium Alloy.
MIL-G-24696 - Gaskets, Sheet, Non-Asbestos.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-438 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
DOD-STD-1399, Section 301 - Interface Standard for Shipboard Systems Ship Motion and Attitude; (Metric).

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

B 214 - Root Connections for Attaching Piping.
803-5959186 - Submarine Heat-Exchanger Anode Plug.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Naval Engineering Drawing Support Activity, Code 202.2, Portsmouth, NH 03804-5000.)

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PUBLICATIONS

NAVSEA
0900-LP-001-7000 - Instructions for Torch Brazing of Ferrous and Non-Ferrous Piping.
0908-LP-000-3010 - Shock Design Criteria for Surface Ships.

(Application for copies should be addressed to the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form). (DOD adopted)
B1.12 - Class 5 Interference-Fit Thread.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 27 - Standard Specification for Steel Castings, Carbon for General Application. (DOD adopted)
A 36 - Standard Specification for Structural Steel, General Application. (DOD adopted)
A 105 - Standard Specification for Forgings, Carbon Steel, for Piping Components. (DOD adopted)
A 106 - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service. (DOD adopted)
A 108 - Standard Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality. (DOD adopted)
A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DOD adopted)
A 181 - Standard Specification for Forgings, Carbon Steel, for General Purpose Piping. (DOD adopted)
A 276 - Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes. (DOD adopted)
A 285 - Standard Specification for Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength. (DOD adopted)
A 351 - Standard Specification for Steel Castings, Austenitic, for High-Temperature Service.
A 516 - Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate and Lower Temperature Service. (DOD adopted)
A 537 - Standard Specification for Pressure Vessel Plates, Heat-Treated, Carbon-Manganese-Silicon Steel. (DOD adopted)

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ASTM (Continued)

A 582 - Standard Specification for Free Machining Stainless and Heat Resisting Steel Bars, Hot Rolled or Cold Finished. (DOD adopted)

A 675 - Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties. (DOD adopted)

A 743 - Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Corrosion Resistant, for General Application. (DOD adopted)

B 111 - Standard Specification for Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock; (Metric). (DOD adopted)

B 127 - Standard Specification for Nickel-Copper Alloy (UNS No. 4400) Plate Sheet, and Strip. (DOD adopted)

B 151 - Standard Specification for Copper-Nickel-Zinc Alloy (Nickel-Silver) and Copper-Nickel Rod and Bar. (DOD adopted)

B 152 - Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar. (DOD adopted)

B 168 - Nickel-Chromium-Iron Alloy (UNS No. 660^0 and 66900) Plate, Sheet and Strip.

B 169 - Standard Specification for Aluminum Bronze Plate, Sheet, Strip and Rolled Bar. (DOD adopted)

B 171 - Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers.

B 265 - Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate.

B 338 - Standard Specification for Seamless and Welded Titanium and Titanium Alloy Tubes for Condensers and Heat Exchangers. (DOD adopted)

B 381 - Standard Specification for Titanium and Titanium Alloy Forgings.

B 443 - Nickel-Chromium-Molybdenum-Columbian Alloy (UNS No. 6625) Plate, Sheet and Strip.

B 564 - Standard Specification for Nickel Alloy Forgings. (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII, Division I.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

GENERAL DYNAMICS, ELECTRIC BOAT DIVISION

3155 - Tubes, Seamless, Titanium, Condenser and Heat Exchanger.

(Application for copies should be addressed to General Dynamics, Electric Boat Division, 75 Eastern Point Road, Groton, CT 06340-4989.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Air ejector assemblies furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.5).

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be of quality best suited for the purpose intended.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Welding and allied processes. Welding and allied processes shall be in accordance with MIL-STD-278. The stress relieving of the inter and after condenser is waived provided the carbon content of carbon steel in welded assemblies does not exceed 0.35 percent. Brazing of piping joints, including root connections, shall be in accordance with NAVSEA 0900-LP-001-7000. Where these joints are 1/2 inch or larger in nominal size, preinserted ring fittings shall be used. Face feed fittings may be used in sizes smaller than 1/2 inch. Use of brazed joints is prohibited where the joints will be inaccessible for repair.

3.2.3 Mercury. Mercury shall not be used for any service (except fluorescent or mercury vapor lighting) in the production or testing of air ejector components or assemblies. Mercury-containing manometers or thermometers shall not be used.

3.2.4 Cadmium. Cadmium plating shall not be used.

3.3 Air ejector assembly types.

3.3.1 Type I air ejector assembly. Type I air ejector assemblies shall serve the main condenser or ship's service or other auxiliary surface condenser (condensers of types I, II, or III of MIL-C-15430).

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3.3.1.1 Alternative combinations. Type I air ejector assemblies shall consist of either of the following two combinations:

(a) Two two-stage air ejector units with a surface type inter and after condenser.
(b) Two two-stage air ejector units, of which the first stage shall be non-condensing, and the second stage shall handle the motive steam from the first stage as well as the non-condensable gases removed by the unit. Only a surface-type after condenser (no inter condenser) shall be provided with these air ejector units.

3.3.2 Type II air ejector assembly. Type II air ejector assemblies shall serve the distiller condenser or the last stage condenser of a distilling unit as specified in MIL-D-18641.

3.3.2.1 Components. Type II air ejector assemblies shall consist of one air ejector unit with a surface-type after condenser. The air ejector unit shall be either single stage or multiple stage as required by the characteristics of the distilling unit which it serves. Two-stage air ejector units shall be configured so as not to require an inter condenser. A separate inter condenser may be provided for a three-stage air ejector unit to condense the steam from the first stage.

3.3.3 Type III air ejector assembly. Type III air ejector assemblies shall serve the condenser provided for the ship's heating drain system or shall be provided for evacuating the non-condensable gases and vapors from the turbine gland leakoff system.

3.3.3.1 Combinations. Type III air ejector assemblies shall consist of a single stage and shall be provided in combination with the ship's heating drain system condenser (type IV of MIL-C-15430) or as a gland exhaust ejector in combination with a type I air ejector assembly.

3.4 Construction. Construction of air ejector assemblies shall be as specified in 3.4.1 through 3.4.17 (see 6.3).

3.4.1 Structural design calculations. Structural design calculations for air ejector assemblies shall be in accordance with ASME Boiler and Pressure Vessel Code, Section VIII, Division I.

3.4.2 Space and weight. Space occupied and weight of air ejector assemblies shall be held to a minimum consistent with the requirements of this specification.

3.4.2.1 Submarine air ejector assembly dimensions. The dimensions of air ejector assemblies to be installed in submarines shall be such that the air ejector assembly or its components can be removed through a standard hatch, or, if the compartment in which the unit is to be installed is hatchless, such that the unit can be moved through bulkhead openings into a compartment that is provided with a hatch. Hatch and bulkhead opening dimensions shall be as specified (see 6.2).

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3.4.3 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4.4 Screw threads. Thread form, dimensions, and tolerances shall be in accordance with ANSI B1.1.

3.4.5 Threaded parts. Threaded parts shall be in accordance with MIL-S-1222 except that materials shall be as specified herein. The construction shall be such that standard wrenches can be used throughout. Tapered pipe threads shall not be used. A class 5 interference fit in accordance with ANSI B1.12 shall be used for assembly of tap-end of studs. Studs shall not be bottomed or shouldered. For the set end of studs, a class 3 fit used with sealing compound, grade AV or AVV of MIL-S-22473, may be substituted for a class 5 interference fit where temperatures do not exceed 220 degrees Fahrenheit (°F).

3.4.5.1 Thread lubricants. Threaded fasteners that have torque requirements shall be lubricated before assembly. The lubricant shall be in accordance with the following:

(a) For fasteners exposed to steam or condensate, the thread lubricant shall be graphite in isopropanol in accordance with MIL-L-24131 (Military Symbol CGI).
(b) For all other applications, the thread lubricant shall conform to one of the following:

(1) Graphite in isopropanol in accordance with MIL-L-24131 (Military Symbol CGI).
(2) Molybdenum disulfide in isopropanol in accordance with MIL-L-24478.
(3) Antiseize compound in accordance with MIL-A-907.

3.4.5.2 Fastener types. Fastener types shall be in the following order of preference:

(a) Through bolt or through (two-nut) stud.
(b) Tap-end stud (one-nut).
(c) Cap screw.

3.4.5.3 Cap screws. Cap screws shall not be used for waterbox-to-shell or waterbox-to-tube sheet bolting, for inspection cover bolting, or for corrosion-preventive anode support cover bolting.

3.4.5.4 Collar or stud bolts. Where collar bolts or stud bolts are used to make up a multiple flange joint, they shall have a square extension beyond the threads on one end for use of a wrench to prevent turning of the bolt when the nuts are tightened or removed.

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3.4.5.5 Thread engagement. Stud engagement shall hold the maximum design load. Engagement shall be not less than the diameter of the fastener. Nuts shall be fully engaged. For a threaded fastener, not less than one thread nor more than four threads shall protrude beyond the crown of the nut. For threaded fasteners that have torque requirements, the bolting shall be tightened to a stipulated torque value by means of a torque wrench.

3.4.5.6 Screw thread selection.

3.4.5.6.1 Unified thread series. Unless otherwise specified herein, screw threads shall be of the unified thread series in accordance with ANSI B1.1.

3.4.5.6.2 Coarse and fine thread series. The coarse thread series shall be used unless the component design indicates a necessity for the fine thread series.

3.4.5.6.3 Eight-thread series. For fasteners of diameter 1 inch and larger, the eight-thread series shall be used.

3.4.6 Flanges, valves, piping, and fittings. Flanges, valves, piping, and fittings shall be in accordance with MIL-STD-777 for surface ships, and MIL-STD-438 for submarines. Root connections shall have the increased wall thickness as specified in Drawing B 214. Nozzles shall be the minimum length possible for installation or removal of through bolts.

3.4.7 Support against ship motion and attitude, and shock.

3.4.7.1 Motion and attitude. Unless otherwise specified (see 6.2), air ejector assemblies shall operate satisfactorily under the ship motion and attitude conditions given in section 301 of DOD-STD-1399.

3.4.7.2 Assembly support. The air ejector assembly shall be supported by plates or brackets such that the primary means of support shall be other than the clamping action of the bolts securing the inter and after condenser shell end, tube sheet, or waterbox flange joints. The air ejector assembly and its components shall be supported by a structure other than nozzle flanges and piping. The air ejector units shall be supported from the inter and after condenser shell by supplemental supports as necessary to secure against damage from shock (see 3.4.8.1).

3.4.8 Shock and vibration.

3.4.8.1 Shock resistant. Shock tests shall be performed in accordance with MIL-S-901 and 4.6.4.1 herein. Shock resistance of air ejector assemblies shall be grade A or B, as specified (see 6.2).

3.4.8.1.1 Shock analysis in lieu of test. Air ejector assemblies that cannot be shock tested shall be dynamically analyzed in accordance with NAVSEA 0908-LP-000-3010. The air ejectors and their sub-bases, feet, structural members, and hold-down bolts shall be included in the analysis.

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3.4.8.1.2 Bolt holes for shear-stressed bolts. Bolts that will be stressed in shear shall be installed in holes not greater in diameter than the following:

Nominal bolt diameter (inches)	Maximum diameter of hole (inches)
3/4 and smaller	Nominal bolt diameter plus 1/32
Larger than 3/4	Nominal bolt diameter plus 1/16

3.4.8.1.3 Rigidity of support structures. Assemblies that are rigidly supported shall be attached only to structures that cannot deflect relative to one another under shock loadings.

3.4.8.1.4 Shock mounts. Shock mount design shall have approval by the contracting activity prior to installation of the shock mounts (see 6.2).

3.4.8.1.5 Deflection snubbers. Where snubbers shall be installed to limit deflection under shock loading, they shall be in accordance with NAVSEA 0908-LP-000-3010.

3.4.8.2 Environmental vibration. Equipment shall be able to withstand environmental vibration, type I of MIL-STD-167-1 over the range specified in 6.2, applied at the foundation. Testing shall be conducted when specified (see 4.7 and 6.2).

3.4.9 Shell expansion. To provide for expansion of the shell (see 3.5.8.3 and 3.6.4.3) the foot at one end of the shell shall allow changes in length of the shell. Acceptable methods shall be as follows:

(a) One foot shall be rigid and, together with its foundation bolts, shall withstand the whole high impact shock loading in the direction parallel to the longitudinal axis of the shell. The other foot, or the member to which it connects, shall flex with changes in length of the shell. Both feet shall be solidly bolted to the foundation.

(b) One foot shall be rigid and, together with its foundation bolts, shall withstand the whole high impact shock load in the direction parallel to the longitudinal axis of the shell. This foot shall be solidly bolted to the foundation. The other foot shall be rigid and shall be provided with slotted holes for foundation bolts. Bushings shall be installed around the foundation bolts and shall be provided with a shoulder of larger diameter than the width of the slotted hole. Clearance between the under surface of the shoulder and a finished or spot-faced surface on the top of the foot shall be provided, but shall not exceed 0.005 inch. Clearance between the foundation bolt and the bore of the bushing and between the bushing and the inner sides of the slotted hole shall be in accordance with 3.4.8.1.2. The foundation bolts shall hold the bushings solidly to the foundation.

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3.4.10 Capacity. Air ejector units shall have sufficient capacity to serve the condenser with which they operate over the entire range of the condenser operation (see 6.2). The choice of capacity shall be the responsibility of the condenser or distilling unit manufacturer, but the corresponding capacity of each air ejector unit of a type I assembly shall be not less than shown in table I.

TABLE I. Capacity of type I condenser.

Maximum steam condensed, pounds per hour	Capacity, cubic feet per minute, 70°F, free dry air
Up to 12,500	1.5
12,501 to 25,000	3.0
25,001 to 50,000	4.0
50,001 to 100,000	5.0
100,001 to 250,000	7.5
250,001 to 500,000	10.0
500,001 and up	12.5

3.4.10.1 Allowance for water vapor. Air ejector units for type I assemblies shall have sufficient capacity to handle the amounts of free dry air specified in table I plus the water vapor to saturate the air. The rated saturation pressure shall be considered as 1 inch of mercury (Hg) absolute or as 1 inch Hg less than the absolute pressure at the steam inlet of the condenser served, based on the full power ahead steam condensed and the design inlet circulating water temperature as specified in MIL-C-15430, whichever resulting absolute pressure is greater. The saturation temperature shall be considered as 7.5°F less than the saturated steam temperature corresponding to the absolute pressure.

3.4.11 Air ejector unit materials. Air ejector units shall be constructed of the material specified in table II.

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TABLE II. Material for air ejector units.

Part	Material	Applicable document
Steam chest	Steel bar or steel bar, grades 1008 thru 1035; or forged steel; or forged steel, class 60; or steel bar, grade 55; or steel plate[1]	ASTM A 36 ASTM A 108 ASTM A 105 ASTM A 181 ASTM A 675 ASTM A 516
Suction chamber	Cast steel, grade 60-30; or fabricated steel, pressure vessel quality; or steel bar, grade 1008-1035; or steel, corrosion resisting, cast, grade CF-8 or CF-8M; or forged steel	ASTM A 27 ASTM A 285 or ASTM A 537 ASTM A 108 ASTM A 743 or ASTM A 351 ASTM A 105
Diffusers	Steel, corrosion resisting, type 304 or 316, bar; or steel, corrosion resisting, cast, grade CF-8 or CF-8M;	ASTM A 276, ASTM A 743
Nozzles	Steel, corrosion resisting, type 304 or 316; or type 303	ASTM A 276 ASTM A 582
Nozzle rings	Steel, corrosion resisting, type 304 or 316; or type 303	ASTM A 276 ASTM A 582
Gaskets (nozzle)	Copper, annealed, C12500; or steel, sheet, hot rolled, commercial; or nickel-copper alloy, soft annealed	ASTM B 152 ASTM B 127
Bolts and nuts	Steel, high temperature service, grade B7	MIL-S-1222

[1]For steam chest covers only.

3.4.12 Steam strainer. A steam strainer shall be provided for each ejector element. Strainers shall be made of material in accordance with table III. Steam supply valves and piping shall not be provided with the air ejectors. Steam flow shall be directed into the inside of the strainer baskets, and convenient bolted cleanout covers shall be provided for removal of the strainer baskets along with the entrapped impurities. Steam outlet holes in strainer baskets shall be not more than 1/16 inch in diameter, and the ratio of the sum of the areas through strainer basket steam outlet holes to the area of the inside diameter of the strainer steam inlet shall be not less than 2.5 to 1. Covers shall not be fitted with any connection for a blow valve. Covers may be straight thread plugs with metal gaskets if threaded size is less than or equal to 1 inch in diameter. An acceptable alternative is to use a separate strainer attached by welding to the steam chest to form a compact arrangement. The separate strainer shall meet these requirements, and the inlet connection shall be provided with a flange.

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TABLE III. Material for strainers.

Part	Material	Applicable document
Baskets	Steel, corrosion resisting, type 304 or 316;	ASTM A 167
	or nickel-copper alloy	ASTM B 127
Castings	Cast steel, grade 60-30;	ASTM A 27
	or bar, grades 1008 thru 1035	ASTM A 108
Forgings	Forged steel, class 60	ASTM A 181

3.4.13 Nozzle quantities and steam conditions. Air ejector units shall have a single steam nozzle in each stage, and shall operate at steam pressures and temperatures as specified (see 6.2).

3.4.14 Air ejector nozzles. Nozzles shall be machined to size, with nominal internal finish of 63 microinches root mean square (rms) or smoother, and free from imperfections. Nozzles having threaded ends for screwing the nozzles into the steam chests shall have flattened sectors for use of a standard wrench in removal.

3.4.15 Air vent. The after condenser shall have an air vent of sufficient size to prevent back pressure that would affect operation of the air ejectors, despite the length of vent piping to its atmospheric discharge. The vent shall be of adequate size to pass the full steam flow from both air ejectors in case of failure of circulating water, without causing the pressure in the inter and after condensers to exceed that for which they are designed.

3.4.16 Compound gauge. The inter condenser section of the shell shall have a connection for a compound gauge. Unless otherwise specified (see 6.2), the compound gauge shall be provided with the air ejector and shall be in accordance with MIL-G-18997.

3.4.17 Life. Air ejector design life shall be as specified (see 6.2).

3.5 Inter, after, and gland exhaust condensers of type I air ejector assemblies. Unless otherwise specified herein, 3.5.1 through 3.5.9 shall apply to both alternative combinations of type I air ejector assemblies (see 3.3.1.1(a) and (b)).

3.5.1 One-shell construction. For light weight and compactness, the inter, after, and gland exhaust condensers, where used, shall be combined in one shell. The entire assembly shall allow both nozzles of the air ejector not in operation to be dismantled for cleaning and repair without interfering with operation of the other air ejector. Inter and after condensers may be subdivided into separate sections to serve each air ejector, or may be provided in single sections with

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isolating valves provided at the first stage discharge and second stage suction. Isolating valves shall be low-pressure gate or angle valves, or may be manifold valves incorporated in the unit, provided that no undue restriction to flow of the air-vapor mixture occurs. Valves shall be fitted with indicators to show whether valves are in the open or shut position.

3.5.2 Relief valves. Where interstage isolating valves are installed, or any possibility of buildup of steam pressure from the steam supply exists, relief valves shall be installed on the inter condenser shell. Relief valves shall be in accordance with type A-1 of MIL-V-20065. Relief valves shall be sized to pass the full steam flow of both first stage nozzles at maximum supply steam pressure.

3.5.3 Gland exhaust condensers. Where a gland exhaust condenser section is provided as part of the after condenser, and a gland exhaust fan is used (see 6.2), the after condenser heat transfer surface shall be increased to accommodate the additional steam and air from the gland exhaust based on maximum worn gland conditions. A vacuum of not less than 5 inches of water shall be maintainable at the turbine leakoff connections at the maximum worn gland condition by the gland exhaust condenser. If a gland exhaust ejector (type III) is specified (see 6.2), it shall be provided as part of the air ejector assembly. The gland exhaust condenser section shall be integral with the after condenser, but subdivided from the after condenser surface, so that the gland exhaust steam is condensed in the gland exhaust section and the air is evacuated by the gland exhaust ejector, and discharged into the after condenser.

3.5.4 Condenser coolant. Only condensate shall be used for circulation through inter, after, and gland exhaust condensers on surface ships. Fresh water shall be used as the coolant in inter, after, and gland exhaust condensers on submarines. Allowable pressure drop and temperature of coolant for calculation purposes shall be as specified (see 6.2).

3.5.5 Condensing heat transfer surface. Sufficient surface shall be provided in the inter, after, and gland exhaust condensers so that both two-stage air ejector units may be operated at rated capacity. There shall be no discharge of free steam from the after condenser vent when both two-stage air ejector units are operating and the gland exhaust condenser is receiving the rated amounts of gland leakoff. Where a gland exhaust air ejector is provided, its motive steam shall also be effectively condensed.

3.5.6 Tubes.

3.5.6.1 Tube shape and size. Inter, after, and gland exhaust condenser tubes shall be straight or U-bent. Copper-nickel tubes shall be 5/8-inch outside diameter (od), number 18 BWG. Titanium tubes shall be 5/8-inch od, number 18 BWG or 9/16-inch od, number 20 BWG.

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3.5.6.2 Straight-tube ordering length. Straight copper-nickel tubes of 4 feet or more in length shall have an ordering tube length in multiples of 6 inches.

3.5.6.3 Tubes expansion. Tubes shall be expanded into the tube sheet holes at both ends, and copper-nickel tubes shall be belled on a 1/2-inch radius to 3/4-inch od at the inlet ends. The inlet ends shall be finished flush with the face of the tube sheet. The outlet ends shall protrude not more than 3/16 inch beyond the face of the tube sheet, but shall not terminate within the tube sheet.

3.5.6.4 U-bent tube bending radius. U-bent copper-nickel tubes shall have a centerline bend radius of not less than 15/16-inch. U-bent titanium tubes shall have a centerline bend radius of not less than 1 1/4 inch.

3.5.6.5 Tube expansion method. Expansion of tubes shall be controlled by means of an automatic tube expander control using a tube expander in accordance with MIL-E-15809. The tube expander shall be adjusted so that expansion of the tube shall stop within 1/8-inch of the inner face of the tube sheet.

3.5.7 Tube sheets.

3.5.7.1 Tube holes. Tube holes in the tube sheets shall be as follows:

(a) For copper-nickel tubes, holes in the tube sheets shall be 0.626-inch diameter, with a tolerance of plus 0.007-inch, with inner edges slightly rounded. The outer edge of holes for inlet ends shall be rounded on a 1/2-inch radius to a diameter of 3/4-inch. Outer edges of holes for outlet ends shall be free of burrs. Holes for both inlet and outlet ends of tubes shall be provided with one groove, 1/16 or 1/8-inch wide, 0.012 inch deep, with the outer edge of the groove 3/8-inch from the outer (water) side of the tube sheet.

(b) For titanium tubes, holes in tube sheets shall be 0.570-inch diameter with a tolerance of plus or minus 0.003 for 9/16-inch od tubes or 0.632-inch diameter with a tolerance of plus or minus 0.003 for 5/8-inch od tubes. Holes shall have two annular grooves 1/8-inch wide by 0.012-inch deep, with not less than 1/8-inch distance between grooves. Tube sheet tube holes shall be rounded on each face to a 1/16-inch radius.

3.5.7.2 Tube sheet thickness. Tube sheets shall have thickness of not less than 3/4-inch.

3.5.7.3 Tube hole spacing. Allowable hole spacing in tube sheets shall be not less than 13/16-inch center-to-center for copper-nickel tubes. For titanium tubes, center-to-center spacing shall be not less than 13/16-inch for 5/8-inch od tubes or not less than 3/4-inch for 9/16-inch od tubes.

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3.5.7.4 Tube sheet attachment to shell. For U-bent tube units with removable bundles, tube sheets shall be fastened to the flanges of the shell by through bolts that extend through the waterbox flanges. Studs or tap bolts shall be used where it is impractical to install through bolts. Collar bolts or stud bolts shall be installed among the through bolts in sufficient number to maintain a tight joint between the tube sheets and shell flanges when the waterboxes are removed. Bolts shall be renewable without removing the waterboxes. For straight tube units, and for U-bent tube units without removable bundles, tube sheets shall be welded to the shell.

3.5.8 Shells.

3.5.8.1 Shell partitioning. The partitions between inter, after, and the gland exhaust condenser and between the sections thereof, if sectioned, shall be welded. The flanges between the shell and shell end cover, and partitions and shell end cover, shall be sufficiently wide for an adequate gasket. Particular attention shall be given to the spacing of the bolting, so that there shall be no leakage between inter, after, and gland exhaust condensers.

3.5.8.2 Impingement protection. Internal baffles or external impingement boxes shall be provided as necessary to protect tubes from direct impingement of incoming steam.

3.5.8.3 Supporting feet. The supporting feet shall allow for expansion of the shell (see 3.4.9).

3.5.8.4 Allowance for tube expansion and contraction. The shell shall be provided with means to allow free expansion and contraction of the tubes. For straight tubes the means shall be an expansion joint in the shell located outboard of one supporting foot of the after condenser. When necessary, the expansion joint shall be protected against transverse high impact shock. U-bent tubes do not require an expansion joint in the shell.

3.5.8.5 Cleanout ports and drain location. Manual cleanout ports shall be provided in each condenser section to allow access to shell internals for manual removal of corrosion products. The inside diameter (id) of circular access ports shall be not less than 4 inches. Each access port, other than circular, shall provide an open area of not less than 12 square inches, with one id being at least 4 inches, and none less than 3 inches. Condenser drains shall be located and connected to minimize plugging due to corrosion products.

3.5.9 Inter and after condenser materials. The materials used in construction inter and after condensers of type I air ejector assemblies shall be in accordance with either table IV or table VI for surface ship units and either table V or table VI for submarine units (see 6.2).

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TABLE IV. Material for inter and after condensers, type I (surface ships only).

Parts	Material	Applicable document
Shells	Steel, or steel, grade DH-36; or seamless steel pipe, grade A	ASTM A 537 MIL-S-22698 ASTM A 106
Waterboxes	Steel, or steel, plate, marine boiler, firebox quality; or seamless steel pipe, grade A; or steel, grade DH-36	ASTM A 537 ASTM A 285 ASTM A 106 MIL-S-22698
Tube sheets	Bronze, aluminum, alloy C61300, or copper-nickel alloy composition 90-10	ASTM B 169 MIL-C-15726
Tubes	Copper-nickel alloy C70600	ASTM B 111[1]
Support and baffle plates	Bronze, aluminum, alloy C61300, or steel; or copper-nickel alloy C70600	ASTM B 169 ASTM A 537 ASTM B 171
Bolts, studs, and nuts in contact with steam or condensate	Nickel alloy grade 400, 405 or 500	MIL-S-1222
Gaskets[2]	Rubber, synthetic, class 1, grade 80, or non-asbestos sheet	MIL-R-6855 MIL-G-24696
Other bolts, studs and nuts	Steel	MIL-S-1222

[1]Except plus tolerance on od 0.000 inch.
[2]See 3.7.

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TABLE V. Material for inter and after condensers, type I (submarines).

Parts	Material	Applicable document
Shells, support plates and baffle plates	Steel, or steel, grade DH-36; or seamless steel pipe, grade A	ASTM A 537 MIL-S-22698 ASTM A 106
Waterboxes	Copper-nickel alloy, composition 70-30	MIL-C-15726
Tube sheets	Copper-nickel alloy, composition 70-30	MIL-C-15726
Tubes	Copper-nickel alloy, composition 70-30	MIL-T-15005
Steam impingement baffles	CRES, types 316 or 304	Commercial
Bolts, studs, and nuts in contact with steam or condensate, or in joints involving pressure vessel integrity	Nickel alloy, grade 400, 405 or 500	MIL-S-1222
Other bolts, studs, and nuts	Steel	MIL-S-1222
Gaskets[1]	Rubber, synthetic, class 1, grade 80 or non-asbestos sheet	MIL-R-6855 MIL-G-24696

[1]See 3.7.

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TABLE VI. Material for inter and after condensers, type I (corrosion-resistant option).

Parts	Material	Applicable document
Shell	NiCrFe alloy 600 plate	ASTM B 168 UNS No. N06600
Tube support plates[1]	NiCrMoCb alloy 625 plate	ASTM B 443
Shell pressure/ velocity reducing configurations, tube protection baffles, non-structural baffles, bars, tubes or screens, and piping and fittings subject to internal or external impingement	Stainless steel, type 316, 316L	Commercial
Waterboxes	NiCrFe plate or Copper Nickel plate	ASTM B 168 UNS No. N06600 MIL-C-15726 Composition 70/30
Tube sheet[1]	NiCrFe alloy 600 plate, or NiCrMbCb alloy 625 plate	ASTM B 168 UNS No. N06600 ASTM B 443 UNS No. N06625
Tubes	Titanium, commercially pure, grade 2 seamless	EB Specification 3155
Bolts, studs and nuts in contact with steam or condensate, or in joints involving pressure vessel integrity	Nickel alloy, grade 400, 405, Or 500	MIL-S-1222
Other bolts, studs, and nuts	Steel	MIL-S-1222
Gasket for waterbox-to-tubesheet joint	O-ring, type I	MIL-G-21610
Gaskets for shell covers	Gasket, sheet or non-asbestos sheet	MIL-G-24696

[1]NiCrMoCb weld wire shall be used for attachment of NiCrMoCb parts to NiCrFe parts.

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3.6 After condensers and inter condensers of type II air ejector assemblies (surface ships and submarine non-submergence pressure).

3.6.1 After condenser materials. Material for after condensers of type II air ejector assemblies shall be class 1, 2, or 3 as specified (see 6.2).

3.6.1.1 Class 1. After condensers of class 1 shall be constructed of class 1 materials as specified in table VII.

3.6.1.2 Class 2. After condensers of class 2 shall be constructed of class 1 materials as specified in table VII, except that copper-nickel shall be alloy composition 90-10.

TABLE VII. Class 1 materials for after condensers, type II.

Parts	Material	Applicable document
Shells	Copper-nickel alloy composition 70-30 or 90-10, or bronze alloy C90300, C90500; or C92200	MIL-C-15726 QQ-C-390
Waterboxes	Copper-nickel alloy, composition 70-30 or bronze alloy C90300, C90500; or C92200	MIL-C-15726 QQ-C-390
Tube sheets	Copper-nickel alloy, composition 70-30	MIL-C-15726
Tubes	Copper-nickel alloy, composition 70-30	MIL-T-15005
Shell internals	Copper-nickel alloy, composition 70-30	MIL-C-15726
Bolts, studs or nuts in contact with sea water, steam, or condensate, or in joints involving pressure vessel integrity	Nickel alloy, grade 400, 405 or 500	MIL-S-1222
Gaskets[1]	Rubber sheet, synthetic, or rubber, synthetic, class 1, grade 80, or non-asbestos sheet	MIL-R-21252 MIL-R-6855 MIL-G-24696
Other bolts, studs and nuts	Steel	MIL-S-1222

[1]See 3.7.

3.6.1.3 Class 3. After condensers of class 3 shall be constructed of class 3 materials as specified in table VIII.

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TABLE VIII. Class 3 materials for after condensers, type II.

Parts	Material	Applicable document
Shells	Titanium, grade 2 or 3 or copper-nickel alloy	ASTM B 265 MIL-C-15726
Waterboxes	Nickel-chromium-molybdenum-columbium alloy (UNS N06625)	ASTM B 564 or MIL-C-24615
	Titanium, grade 2 or 3	ASTM B 265 or ASTM B 381
	Copper-Nickel, composition 70-30	MIL-C-15726 or MIL-C-20159
Tube sheets, support plates, and baffles	Titanium, grade 2 or 3	ASTM B 265
Tubes	Titanium, grade 2	ASTM B 338[1]
Bolts, studs, and nuts in contact with sea water, steam, or condensate, or in joints involving pressure vessel integrity	Nickel alloy, grades 400, 405 or 500	MIL-S-1222
Gaskets[2]	Rubber sheet, synthetic, or rubber synthetic class 1, grade 80, or non-asbestos sheet	MIL-R-21252 MIL-R-6855 MIL-G-24696
Other bolts, studs and nuts	Steel	MIL-S-1222

[1]For submarine application, tubes shall be in accordance with Electric Boat Spec 3155.
[2]See 3.7.

3.6.1.4 Inter condenser materials. Separate inter condensers for multiple stage type II air ejectors (see 3.3.2.1) shall be constructed of materials specified in 3.6.1.

3.6.2 Circulating medium. Seawater shall be used for circulation in the after condenser. For submarine application, the seawater shall be non-submergence pressure.

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3.6.3 Tubes and tube sheets.

3.6.3.1 Tube size and length. Copper-nickel tubes shall be 5/8-inch or 3/4-inch od by 18 BWG. Straight copper-nickel tubes longer than 4 feet shall have ordering lengths in multiples of 6 inches. Titanium tubes shall be 9/16-inch od by 20 BWG unless otherwise specified (see 6.2).

3.6.3.2 Tube shape. Tubes shall be straight or U-bent.

3.6.3.3 Tube bending radius. U-bent copper-nickel tubes shall have a centerline radius of not less than 15/16-inch, for 5/8 inch od tubes or 1-1/8 inch for 3/4 inch od tubes. U-bent titanium tubes shall have a centerline radius of not less than 1-1/4 inches.

3.6.3.4 Tube sheet configuration. The after condensers for saturated steam propelled surface ships and submarines shall be provided with double tube sheets at each end of the tube. The double tube sheets shall be constructed with a space of not less than 1/2 inch between them by machining one face of either or both tube sheets, or by using a spacer ring. The space between tube sheets shall have a vent and drain to the atmosphere. The joint between tube sheets or between tube sheet and spacer ring shall be welded. The after condensers for all other applications shall be provided with a single tube sheet at each end of the tube.

3.6.3.5 Tube sheet thickness. Single tube sheets and outer tube sheets of double tube sheet construction shall have a thickness of not less than 3/4 inch. Inner tube sheets of double tube sheet construction shall have a thickness of not less than 7/8 inch.

3.6.3.6 Minimum tube spacing. For copper-nickel tubes, center-to-center spacing shall be not less than 13/16 inch. For titanium tubes, center-to-center spacing shall be not less than 3/4-inch unless otherwise specified (see 6.2).

3.6.3.7 Tube holes. For copper-nickel tubes, holes in tube sheets shall be 0.626-inch diameter for 5/8 inch od tubes or 0.751 inch diameter for 3/4 inch od tubes with a tolerance of plus 0.007 inch and shall have one annular groove, 1/16 or 1/8 inch wide by 0.012-inch deep, located 3/8-inch from the outer (circulating water) side of the tube sheet. Holes in inner tube sheets of double tube sheet construction shall not be provided with a groove. Outer edges of the holes for the inlet tube ends shall be rounded on 1/2-inch radius to 3/4-inch od. Outer edges of the holes for outlet tube ends shall be free from burrs. Inner edges of tube holes shall be rounded to a 1/16-inch radius. For titanium tubes, holes in tube sheets shall be 0.571-inch diameter with a tolerance of plus or minus 0.002 inch unless otherwise specified (see 6.2) and shall have two annular grooves 1/8-inch wide by 0.012-inch deep, with not less than 1/8-inch distance between grooves. Tube sheet tube holes in inner tube sheets of double tube sheet construction and in titanium tube sheets shall be rounded on each face to a 1/16-inch radius.

3.6.3.8 Tube sheet attachment to shell. Tube sheets shall be fastened to the shell as specified in 3.5.7.4.

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3.6.3.9 Tube expansion. Tubes shall be expanded at each end into the tube sheet tube holes by means of an automatic tube expander control and with tube expanders in accordance with MIL-E-15809. In double tube sheet construction, expansion of the tubes in the inner tube sheet holes shall start 1/8 inch inside the outer face and stop within 1/8 inch of the inner face of the tube sheet. In the outer tube sheets of double tube sheet construction and in the tube sheets of single tube sheet construction, the tube expander shall be adjusted so that expansion of the tube shall stop within 1/8 inch of the inner face of the tube sheet. Inlet ends of copper-nickel tubes shall be belled on a 1/2-inch radius to 3/4-inch od, and the ends shall be finished flush with the face of the tube sheet. Inlet ends of titanium tubes shall not be belled. Outlet ends of tubes shall protrude not more than 3/16 inch beyond the face of the tube sheet, but shall not terminate within the tube sheet. Depth of expansion in each tube sheet shall be not less than 5/8 inch.

3.6.4 Shells.

3.6.4.1 Impingement protection. Internal baffles or external impingement boxes shall be fitted in the shell as necessary to protect tubes from direct impingement of incoming steam.

3.6.4.2 Allowance for tube expansion and contraction. Allowance for tube expansion and contraction shall be as specified in 3.5.8.4.

3.6.4.3 Allowance for shell expansion and contraction. The supporting feet shall allow for expansion and contraction of the shell (see 3.4.9).

3.6.5 Corrosion preventive. Waterboxes shall be provided with anodes in accordance with MIL-A-19521 as follows:

(a) For submarine non-submergence pressure applications, bolted cover plates or threaded plugs shall be used for support of anodes. Threaded plugs shall conform to Drawing 803-5959186.
(b) For surface ship applications, support plugs shall be of the straight thread O-ring seal type in accordance with MIL-A-19521. In exception to the material requirements of MIL-A-19521, plug material shall be 70-30 Copper-nickel alloy, MIL-C-15726 or C71500 of ASTM B 151.
(c) If tubes and tube sheets are of titanium, anode support plugs or covers shall be of the same material as the waterbox material if the attached piping is copper-nickel (see 6.2). If the attached piping and waterboxes are of titanium or nickel-chromium-molybdenum-columbium alloy, cathodic protection shall not be provided.

3.7 Flat gaskets. Flat gaskets for condensers shall be as specified in tables IV through VIII, except that where the temperature of metal surfaces in contact with the gasket material may exceed 250°F under any condition of operation, a non-asbestos material in accordance with MIL-G-24696 shall be used. Flat gaskets for condensers shall be 1/16 or 1/32-inch thick. Gaskets that are not replaceable without destructive disassembly shall not be used.

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3.8 Identification plates. Identification plates shall be fastened on each ejector unit. They shall be type A or type B in accordance with MIL-P-15024 and MIL-P-15024/5. Identification plates made of plastic or aluminum are not acceptable.

3.8.1 Securing of identification plates. Identification plates shall be fastened to equipment with corrosion-resistant metallic screws or rivets.

3.8.2 Identification plate data. Ejector unit identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Government contract number.
(c) Command or agency stock number (allow 17 spaces).
(d) Date of manufacture.
(e) Manufacturer's model or type and size.
(f) Service application.
(g) Test pressure, ejector parts (omit if classified).
(h) Test pressure, shell side.
(i) Test pressure, tube side (surface ships only).
(j) Blank space for Government inspector's stamp.
(k) Blank space for unit number (2 spaces - to be stamped by shipyard).

3.9 Painting. Nonferrous castings or parts need not be painted. External surfaces of ferrous metal parts shall be thoroughly cleaned and coated with two coats of heat-resisting paint in accordance with TT-P-28 or equivalent. Interior surfaces of parts shall not be painted or coated with preservative.

3.10 Special tools. Special wrenches or tools needed for service maintenance, including one tube expander for expanding tubes in double tube sheets of after condensers, shall be provided on the basis of one set for each size air ejector assembly provided for each ship (see 6.3). Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification tests. Qualification tests on air ejectors shall be conducted at a laboratory approved by NAVSEA. These tests shall consist of the tests specified in 4.5 through 4.5.4 and 4.6.1 through 4.6.3.3. Air ejectors offered for qualification tests shall be type I. Qualification approval of the type I air ejector shall constitute approval of the type II and III air ejectors.

4.4 Quality conformance inspection. A quality conformance inspection shall be performed that shall include the examination and tests specified in 4.6 (see 6.3).

4.4.1 Inspection lot. A lot shall consist of all air ejector assemblies offered for delivery at one time.

4.4.2 Sampling for quality conformance inspection. One air ejector assembly from each lot shall be subjected to the quality conformance inspection specified in 4.6.2.

4.5 Qualification tests.

4.5.1 Standard test conditions. The standard test conditions for qualification shall be as follows:

(a) Steam supply, pressure 150 pounds per square inch (lb/in^2) gauge, dry and saturated.
(b) Inlet circulating water temperature to inter condenser and after condenser, 90°F.
(c) Mass flow of circulating water to inter and after condenser, 35 pounds of circulating water per pound of steam consumption.
(d) Air-handling capacities and steam consumption obtained for the lower absolute pressure range for saturated air suctions to ejectors at both 80 and 90°F.

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4.5.2 Criteria. Type I air ejectors that undergo tests shall be considered unsatisfactory if their economy, as expressed by pounds of steam per pound of air, exceeds any of the values for the temperatures and pressures shown below:

Saturation temperature (°F)	Absolute pressure (inches Hg)	Economy (pounds steam per pound of air)
90	1.5	56
90	2.0	10
90	2.5	7.5
80	1.5	11.1
80	2.0	7.1
80	12.5	6.3

4.5.3 Performance curve. The air ejectors shall have a characteristic performance curve such that the capacity for air removal shall vary with saturated air-vapor temperature as shown below. The figure in the column entitled "capacity ratio" is the minimum acceptable ratio of capacity for air removal at first-stage suction pressure to the capacity at reference pressure.

Saturated air-vapor temperature (°F)	Absolute pressure at first-stage suction (inches Hg)		Capacity ratio
	Reference	Actual	
90	2	2.5	1.25
90	2	3.0	1.35
90	2	4.0	1.55
80	1.5	2.0	1.10
80	1.5	2.5	1.35
80	1.5	3.0	1.45
80	1.5	4.0	1.60

4.5.4 Shock tests. Shock tests for qualification of complete air ejector assemblies shall be performed in accordance with MIL-S-901, with the following modifications. Under the test procedure for medium weight equipment, the total number of blows shall be increased to nine. The nine blows shall consist of three groups of three blows each. For each group, the height of hammer drop and the initial up travel of the anvil table shall be as shown in table I of MIL-S-901.

(a) If feasible within weight limitations of the medium machine, one blow of each group shall be applied parallel to each of three principal axes of the equipment.
(b) If adapters to permit the procedure of 4.5.4 might result in exceeding machine limitations, each group shall comprise the following:

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(1) The first three blows shall be applied with the unit on the standard (horizontal) mounting adapter.
(2) The second three blows shall be applied with the equipment rotated 30 degrees around its principal axis (using a 30-degree adapter).
(3) The third three blows shall be applied with the equipment rotated to incline the principal axis 30 degrees to the horizontal (using a 30-degree adapter).

4.6 Quality conformance.

4.6.1 Examination. Each air ejector assembly offered for delivery shall be examined to determine conformance to this specification regarding design, material, arrangement, construction, dimensions, fit, finish, and workmanship. Parts and subassemblies shall be examined before final assembly for those features that cannot be examined after assembly, but are considered important to satisfactory service use.

4.6.2 Performance tests before shipment. One air ejector assembly for a given design (model) shall be tested for proper operation and performance before shipment. Duplicate units need only be tested with dry air to demonstrate comparable performance. When adequate demonstration of performance has been shown on prior duplicate units, a proposal for dimensional verification and documentation of critical dimensions in lieu of testing shall be considered.

4.6.2.1 Performance test. The air ejector assemblies shall be tested as follows:

(a) Each air ejector unit shall be operated for determination of capacity (pounds per hour free dry air plus vapor to saturate) versus absolute pressure (inches Hg). The suction-side pressure during test shall range from shutoff pressure to an absolute pressure of 2 inches Hg greater than that for the rated capacity point. The air ejector unit shall demonstrate operation throughout the entire range of its rated capacity.
(b) Air ejector units for submarines shall start and operate against the maximum ambient pressure in the hull (see 6.2).
(c) Steam consumption at the specified inlet steam pressure and temperature shall be determined by weighing the condensate from the inter and after condenser or by calculation from the measured diameter of the nozzle throat.

4.6.2.2 Spare nozzles. Nozzles provided as repair parts shall not be tested in the ejectors with which they are provided. Measurements shall be taken to ensure that the spare nozzles are of the same physical dimensions as the service nozzles.

4.6.3 Hydrostatic tests. Hydrostatic tests shall be conducted in accordance with 4.6.3.1 through 4.6.3.3.

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4.6.3.1 Steam chests and strainers. The steam chests and steam strainers of air ejectors shall be tested at a hydrostatic pressure of 1.5 times the working pressure, except that for air ejectors for saturated steam propelled surface ships and submarines, the test pressure shall be 1.5 times the full generated steam pressure. Other air ejector parts shall be tested hydrostatically to 50 lb/in^2 gauge.

4.6.3.2 Condenser shell sections. The inter condenser, after condenser, and gland exhaust condenser shell sections shall be separately tested to a pressure of 50 lb/in^2 gauge, and leakage observed in the welds or across the shell partitions between sections shall be corrected. This test can be performed either prior to installation of tubes or after installation of tubes, provided leakage across the shell partitions can be observed and corrected.

4.6.3.3 Condenser tubes. After installation of tubes, the following hydrostatic tests shall be applied:

(a) Type I inter, after, and gland exhaust condensers and type II after condensers for surface ships (except saturated steam propelled) shall have:

(1) A 50 lb/in^2 gauge pressure test applied in the shell to demonstrate lack of leakage from individual tubes due to defective tubes or imperfect tube end expansion.

(2) After the test specified in 4.6.3.3(a)(1), the water side shall be tested to a pressure of at least one and one-half times the design pressure of the ship's condensate or fresh water cooling system as specified (see 6.2), but not less than 50 lb/in^2 gauge. Leakage in waterboxes or gasketed joints shall be corrected.

(3) Satisfactory completion of the tests specified in 4.6.3.2, 4.6.3.3(a)(1) and 4.6.3.3(a)(2) as applicable shall be demonstrated following any additional tube expansion or tube replacement, weld repair, gasket replacement, etc., required by the afore-mentioned tests.

(b) Type II after condensers for saturated steam propelled surface ships and submarines shall undergo the following tests:

(1) A 50 lb/in^2 gauge test shall be applied in the shell to demonstrate that no defective tubes are installed. The drains from the spaces between the double tube sheets shall be examined for evidence of leakage between the expanded tube ends and the walls of the tube holes in the inner tube sheets. Leakage discovered shall be corrected by minimal rerolling of the tubes into the inner tube sheets.

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(2) After the test specified in 4.6.3.3(b)(1), the spaces between the double tube sheets shall be tested to 50 lb/in^2 gauge. The pressure shall then be increased to not less than twice the maximum designed discharge pressure of the pump supplying cooling water to the after condenser as specified (see 6.2); and the expanded tube ends in the outer tube sheets (waterbox side) shall be examined for leakage. Leakage shall be corrected by minimal rerolling of the leaking tubes.

(3) The same pressure specified in 4.6.3.3(b)(2) shall then be applied to the seawater side, and leakage in waterboxes or gasketed joints corrected.

4.6.4 Shock resistance.

4.6.4.1 Shock tests. Shock tests shall be performed as specified in 4.5.4 (see 6.3).

4.6.4.2 Shock test extensions. Shock test extensions shall be considered by the command or agency concerned for assemblies that are of similar design, construction, weight, and material to units that have successfully passed the tests specified in 4.5.4. Shock test extension requests shall be prepared in accordance with the instructions specified in MIL-S-901.

4.7 Environmental vibration testing. The air ejector assembly shall be tested for MIL-STD-167-1 environmental vibration, when specified (see 6.2 and 6.3).

4.8 Tests. When an air ejector is to be furnished as part of a main propulsion unit, if the main propulsion unit is tested, the air ejector shall also be tested as part of the propulsion unit. The contractor shall correct any deficiencies which are found during the test.

4.9 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8.)

5.1 Packaging requirements. Air ejector assemblies and accessories shall be preserved level A, C or commercial, packed level A, B, C, or commercial, and marked in accordance with MIL-T-17286, as specified (see 6.2), and shall include bar codes and applicable packaging acquisition options stated therein. In addition, for Navy acquisitions, the following applies:

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(a) Navy fire-retardant requirements.

(1) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant
Category 1 - general use

Level C - Type I - non-weather resistant
Category 1 - general use

(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density, requirements of PPP-F-320.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The air ejector assemblies described in this specification are intended for service of steam condensers on board Navy surface ships and submarines. The three types of air ejector assemblies fit their respective condenser applications as shown in 1.2 and 3.3.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Specific limitations as to weight, space, foundations, sizes, and relative positions of all connecting flanges, the maximum permissible steam consumption, and other necessary restrictions where any such limitations are essential to the installation in question. This information should be shown on a space diagram.
(d) Type and class required (see 1.2).
(e) Hatch and bulkhead opening dimensions (see 3.4.2.1).
(f) Ship motion and attitude requirements, if different from those of DOD-STD-1399, section 301 (see 3.4.7.1).
(g) Shock resistance grade required (see 3.4.8.1).
(h) Shock mount design (see 3.4.8.1.4).
(i) Range of environmental vibration (see 3.4.8.2) and if environmental vibration testing is required (see 4.7).

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(j) Air removal capacity of each ejector, (given in pounds of free dry air per hour, plus associated vapors) at designated absolute pressure and temperature (see 3.4.10).
(k) Supply and operating steam pressures and temperatures (see 3.4.13), and maximum ambient pressure in the hull (see 4.6.2.1(b)), air ejector units for submarines only).
(l) Whether an inter condenser pressure gauge (compound gauge) is not to be provided (see 3.4.16).
(m) Intended design life (see 3.4.17).
(n) Whether a gland exhaust fan will be used or whether a gland exhaust ejector is to be provided as part of the air ejector assembly, and amounts of turbine gland leakoff steam and air and vacuum required (see 3.5.3).
(o) Allowable pressure drop and temperature of coolant through inter, after, and gland exhaust condensers for designated quantity of condensate circulating water (see 3.5.4).
(p) Coolant medium and temperature of circulating water for supply to inter and after condensers (see 3.5.4).
(q) Specify either table IV, table V, or table VI materials for type I air ejector condensers (see 3.5.9).
(r) If type II is required, specify class of material for after condensers (see 3.6.1).
(s) For titanium tubes, the outside diameter and wall thickness required if other than specified (see 3.6.3.1).
(t) For titanium tubes, the minimum tube spacing if other than specified (see 3.6.3.6).
(u) For titanium tubes, the tube hole requirements if other than specified (see 3.6.3.7).
(v) For type II assemblies of class 3 materials, specify what the attached piping material will be (see 3.6.5(c)).
(w) Design pressure of ship's condensate or fresh water cooling system (see 4.6.3.3(a)(2)).
(x) Maximum designed discharge pressure of the pump supplying circulating water to the type II after condenser on saturated steam propelled surface ships and submarines (see 4.6.3.3(b)(2)).
(y) Level of preservation and packing required (see 5.1).
(z) Fire-retardant material requirements (see 5.1(a)).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.4 and appendix A	DI-DRPR-80651	Engineering drawings	----
3.10	DI-V-7007	Tools and test equipment list	----
4.1.1	DI-QCIC-81110	Inspection and test plan	----
4.4	DI-MISC-80678	Certificate/data report	----
4.6.4.1	DI-ENVR-80707	Shock test report	----
4.7	UDI-T-23762	Report, vibration testing	----
4.5 and 4.6.2	DI-MISC-80653	Test report	----

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Special guidance when ordering equipment for the Naval Sea Systems Command (NAVSEA). The table below lists, as a separate line item, every candidate data deliverable arising from this specification and lists the reference paragraph for the deliverable. When this specification is invoked in a contract to purchase equipment for NAVSEA, items marked with a "Yes" should be prepared and delivered:

Reference paragraph	Name of deliverable Name of deliverable	Desired on orders for equipment for NAVSEA
3.4 and appendix A	Drawings	Yes
4.5	Report of qualification tests	Yes
4.6.2	Report of performance tests	Yes
4.6.4	Shock test report	Yes
4.7	Report of environmental vibration test	Yes
6.4	Technical manual	Yes

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. 15465 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Commander, SEA 05Q22, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests must be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.

6.6 Cross-reference of ejector types. Air ejector assembly types I, II, and III were formerly known (before 1986) as types A, B, and C, respectively. This change in designation does not affect storability or usability. Old stocks need not be altered merely to change type designation.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.7.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Torquing procedures. A full description of torquing procedures for bolting should be contained in the technical manual.

6.10 Subject term (key word) listing.

Condensers
Heat exchangers
Jet pumps
Steam condensers
Vacuum distillation

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6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4320-N273)

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APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. The types of drawings required are as follows:

30.1.1 External arrangement drawings. An external arrangement drawing, also called an outline drawing, shall show necessary external views of the air ejector assembly. The outline drawing shall include the external dimensions required for reproduction on ship's machinery arrangement drawings and for guidance in designing the foundation structure for the assembly. The drawing dimensions shall also provide guidance for installation on board ship, including connection of the assembly to external piping. The drawing shall show the space required for removal and replacement of tubes in the heat exchanger components, and for removal of waterboxes, access covers, and corrosion preventive anodes. The drawing shall also show the location of lifting lugs or eyebolts for handling the unit or its components, and the size of openings to be provided for lifting. Drawings shall contain instructions for torquing of bolts.

30.1.2 Assembly drawing. An assembly drawing shall show complete longitudinal and transverse cross-sectional views of the air ejector assembly. This drawing shall show the relationship of parts, arrangement of tubes in heat exchanger components, the method of allowing for tube and shell expansion, arrangement of baffles, and welding of adjacent parts. Enlarged views or sections shall be used in sufficient quantity to show adequate detail. Subassembly drawings conforming to the foregoing may be furnished for individual components of the unit. The drawing or drawings shall be such that a thorough understanding of the design and construction of the apparatus may be obtained without reference to related detail drawings.

30.1.3 List of materials. The assembly drawing shall contain a list of materials showing names of parts with identifying numbers and materials. The identifying numbers shall also be shown adjacent to their respective parts depicted in the various views, with arrows pointing to the parts.

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30.1.4 Detail drawings. Detail drawings shall be made for major parts such as shells, waterboxes, tube sheets, tube supports, baffles, and air ejector elements. The drawings shall be sufficiently dimensioned, with finishes and welding symbols indicated, to allow their manufacture. Detail drawings need not be submitted for approval if sufficient information is provided on the assembly drawings. Detail drawings will, however, be required as part of the final drawings.

30.1.5 Certification data sheets. Certification data sheets shall be furnished and shall include a tabulation of the following data:

(a) Pressure and superheat temperature of main supply steam.
(b) Operating pressure and superheat temperature of steam supply to nozzles after passing through pressure-reducing valve.
(c) Steam consumption for each stage, in pounds mass per hour.
(d) Total steam consumption for each ejector at the specified steam pressure and superheat conditions, in pounds mass per hour.
(e) Air-handling capacity as rated in pounds of dry free air per hour plus associated or entrained vapors at operating absolute pressures and air suction temperatures for at least three separate conditions in the usual operating range.
(f) Cooling surface area of inter condensers.
(g) Cooling surface areas of after and gland exhaust steam condensers.
(h) Mass flow rate of cooling water, in pounds per hour, at the expected temperature necessary for satisfactory operation, and the resultant temperature rise of cooling water.
(i) Pressure drop through inter and after condensers with the maximum specified mass flow rate of circulating water through inter and after condensers.
(j) Maximum permissible temperature of discharge cooling water from inter and after condenser for satisfactory operation, together with the minimum mass flow rate of condensate cooling water necessary.
(k) Dry, operating and flooded weight of assembly.
(l) Hydrostatic test pressures for pressure parts (except for type I after condensers for submarines; for them, reference shall be made to confidential shipbuilder's drawing that gives hydrostatic test pressures for parts subjected to submergence pressure.)
(m) List of repair parts furnished.
(n) Number, wall thickness, and ordering length of tubes.

30.1.6 Intended use. Drawings shall be sufficient to:

(a) Provide design information sufficient to assure conformance to the requirements of this specification and compatibility with the ship and with ship's systems.
(b) Enable evaluation of the suitability of the design for intended naval use.
(c) Enable evaluation of performance and maintenance capability.
(d) Enable shipyard installation without contractor assistance.
(e) Enable ship and shore activities to repair and maintain the apparatus.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-E-15465H(SH)	930524

3. DOCUMENT TITLE
EJECTOR ASSEMBLIES, AIR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL POINT OF CONTACT: Mr. Walter Aerni, SEA 05X23	b. TELEPHONE (Include Area Code) (1) Commercial: 703-602-6794 (2) AUTOVON: 332-6794
ADDRESS (Include Zip Code) COMMANDER, SEA 05Q42 NAVAL SEA SYSTEMS COMMAND 2531 JEFFERSON DAVIS HWY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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